

Functions

Functions (such as `pow`) are called using *call expressions* (such as `pow(10, 3)` which returns 1000; 10 to the third power). The body of a function can `print` or `return` values (or both).

- When `print` is called anywhere, its arguments are displayed right away.
- Executing a return statement stops the function call and provides the value for its call expression.
- When the end of a function body is reached without a `return`, the function returns `None`.

Tip: If a value needs to be used by code outside of a function, then return the value (instead of printing it).

Q1: Multiply

Implement `multiply`, which takes two numbers `x` and `y`. It prints out an equation showing the result of multiplying `x` and `y`, then returns the result.

Try to multiply numbers together just once in the body of `multiply` so that the Python interpreter doesn't have to perform multiplication more *times* than necessary (pun intended).

```
def multiply(x, y):
    """Multiply x by y and print what happened.

    >>> multiply(3, multiply(multiply(4, 5), 6))
    4 * 5 = 20
    20 * 6 = 120
    3 * 120 = 360
    360
    """
    z = x * y
    print(x, '*', y, '=', z)
    return z
```

Calling `print` on multiple arguments displays those values separated by spaces. For example, the call expression `print(1, '+', 2, '=', 3)` displays `1 + 2 = 3`.

Digits

Each digit of a positive integer corresponds to a power of 10. Here are some examples of manipulating digits, illustrated with `n = 357` and `d = 9`. - Remove the last digit of `n`: `n // 10` is 35. - Remove the last two digits of `n`: `n // pow(10, 2)` is 3. - Add together the last two digits of `n`: `n % 10 + n // 10 % 10` is 12. - Put digit `d` at the end of integer `n`: `n * 10 + d` is 3579.

Q2: Cut One Out

Implement `cut`, which takes non-negative integers `n` and `k` and has only a return statement in its body. It returns a positive integer with all of the digits of `n` except the digit that is `k` to the left of the rightmost digit (the one's digit).

2 Functions

If k is 0, then it returns n without its one's digit. If there is no digit k to the right of the one's digit, then it returns n .

```
def cut(n, k):  
    """Return n with the kth digit from the right removed.  
  
    >>> cut(3579, 2)  
    379  
    >>> cut(3579, 0)  
    357  
    >>> cut(3579, 1)  
    359  
    >>> cut(3579, 5)  
    3579  
    """  
    return n // pow(10, k + 1) * pow(10, k) + n % pow(10, k)
```

Please don't look at the hint until everyone in your group agrees that you're stuck and need some extra help.

To implement a function, it can be helpful to open a Python interpreter and focus on an example. Here are all the pieces you need to put together to solve this problem for the example $n=3579$ and $k=2$.

```
>>> n = 3579  
>>> k = 2  
>>> pow(10, k)  
100  
>>> pow(10, k + 1)  
1000  
>>> n // 1000  
3  
>>> 3 * 100  
300  
>>> 3579 % 100  
79  
>>> 300 + 79  
379
```